

The Two-Wheeler Blind Spot

Sizing India's motor insurance market when nearly half the vehicles on the road aren't actually covered.

A new case for the guesstimate shelf — built because BFSI sizing questions are showing up in first rounds more often than most prep material accounts for, and none of them look like the standard cabs-and-fridges problems.

Before You Start

Every guesstimate that's been passed around the batch so far has been about counting things that visibly exist — cabs on a street, fridges in a home, credit cards in a wallet. This one is different. It's about counting something that, for more than half the population in question, isn't actually there.

Insurance sizing questions have been turning up in first and second rounds more than they used to, especially with boutique firms that work with NBFCs and insurers. It's a good test precisely because the "obvious" version of the answer is wrong. Most people size this market by assuming every registered vehicle carries a policy. It doesn't. Roughly 55–60% of two-wheelers in India ride around without valid insurance at any given time, and that single fact changes the entire shape of the estimate.

Work through this the same way you'd work through the operational-risk case on the metals industry already in the folder: segment first using a variable that actually drives different behaviour, apply assumptions inside each segment instead of at the top, and then sanity-check the output against whatever real number you can dig up — even a rough one. That habit matters more than getting any single figure exactly right.

THE CASE PROMPT

"One of our clients is a digital-first general insurance company. They currently sell health and travel cover, and they're deciding whether to enter two-wheeler insurance in India. Before we commission full diligence, I want your back-of-the-envelope estimate of the annual Gross Written Premium for two-wheeler insurance in India today. Talk me through your thinking as you go."

You have a blank sheet, a pen, and roughly 15 minutes. No laptop, no search engine — just what you already know or can reasonably assume.

Clarifying Questions Worth Asking First

- **Scope of the market** — the entire India two-wheeler insurance industry (all insurers), or just the addressable opportunity for one new entrant?
- **Type of cover** — mandatory third-party (TP) only, or TP plus comprehensive/own-damage (OD) policies combined?
- **Commercial fleets** — are delivery-rider two-wheelers (Swiggy, Zomato, Amazon last-mile) in scope, or personal-use vehicles only?
- **Flow vs. stock** — GWP is this year's premium income (mostly renewals plus new policies), not the total value of vehicles on the road. Worth stating out loud so the interviewer knows you know the difference.

For a 15-minute guesstimate, it's fine to say: "I'll scope this to personal-use two-wheelers, combine TP and comprehensive, and treat EVs and petrol bikes together — happy to peel those apart if useful," and then just move.

Building the Number

1 Start with the base — how many two-wheelers are actually on the road

Not how many were sold this year. How many exist, full stop. India registers close to 2 crore new two-wheelers a year now (1.89 crore in 2024, just over 2 crore in 2025), and has been doing so at scale for over a decade. Government VAHAN data and independent trackers converge on a total operating parc somewhere in the 26–30 crore range as of the mid-2020s.

Working assumption: ~28 crore two-wheelers on Indian roads today — the single largest two-wheeler fleet of any country on earth. Nobody expects you to know this to the decimal; what they're checking is whether your order of magnitude is sane and whether you can defend it if pushed.

2 Segment by engine capacity — not by geography

The instinct in most guesstimates is to split urban vs. rural first. Resist it here. The regulator itself (IRDAI) prices third-party premiums by engine capacity band, and that band happens to correlate tightly with income, financing behaviour, and — as you'll see in Step 3 — how likely someone is to actually keep their insurance current. So segment on the variable the rest of your logic will hang off.

Segment	What it actually is	Share of parc	Population (crore)
≤75cc	Legacy mopeds, low-power commuter bikes	5%	1.40
75–150cc	Mass commuter — Splendor, Activa, Pulsar 125	78%	21.84
150–350cc	Premium commuter / cruiser — Classic 350, Duke	15%	4.20
>350cc	Superbikes and premium tourers	2%	0.56
Total		100%	28.00

These shares are a defensible guess based on how India's two-wheeler market is known to be dominated by sub-150cc commuter bikes — not a cited statistic. Say so if asked.

3 Apply insurance penetration — the step every generic casebook skips

This is the crux of the whole case. IRDAI's own data, echoed by a widely-cited General Insurance Council survey, puts uninsured vehicles at roughly 55–60% of everything on Indian roads — and two-wheelers make up the overwhelming majority of that uninsured pool. People buy a bike, insure it for the mandatory long-term period at purchase, and then simply let the policy lapse at renewal. There's no roadside enforcement mechanism that catches this at scale, so the leak just sits there.

REALITY CHECK

If you didn't know this stat walking in, you're not expected to. The move is to say out loud: "I'd guess a meaningful share of two-wheelers are uninsured despite the legal mandate — say 30–40% — because enforcement is weak and renewal reminders are easy to ignore." That's a defensible assumption even without the exact number. Getting the *direction* right (penetration is well under 100%) is what separates a strong answer from a forgettable one.

Penetration isn't uniform across segments either — cheaper, older bikes skew uninsured (low residual value, cash-purchased, no lender forcing renewal); premium bikes skew insured (financed, higher owner engagement, dealership push).

Segment	Population (cr)	Assumed penetration	Insured (cr)
≤75cc	1.40	25%	0.35
75–150cc	21.84	40%	8.74
150–350cc	4.20	65%	2.73
>350cc	0.56	90%	0.50
Total	28.00	44% blended	12.32

Where the insurance gap actually sits, by engine-capacity segment

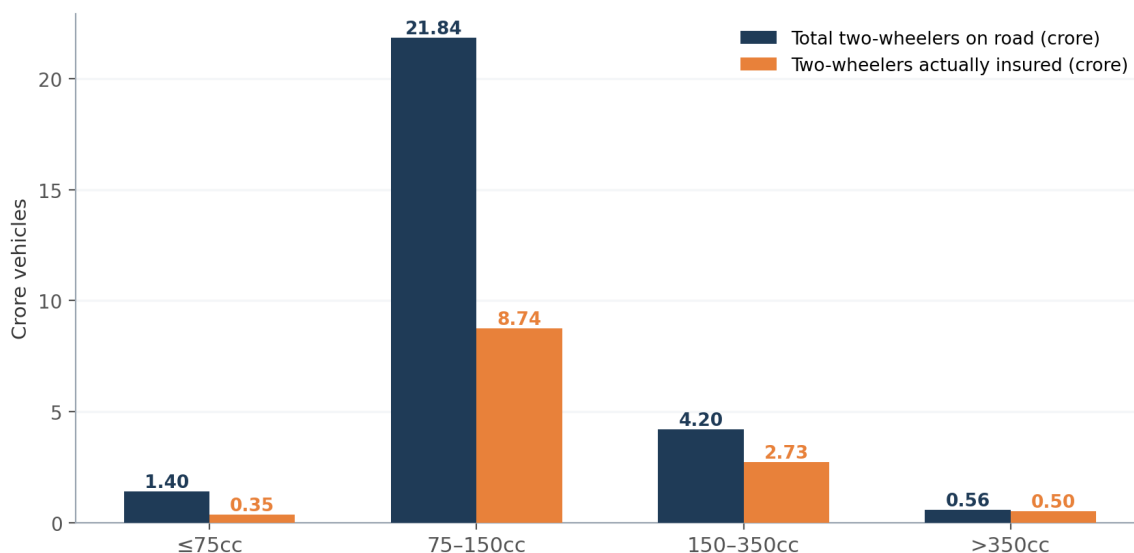


Fig. 1 — The gap is concentrated in the mass commuter segment, in both absolute and relative terms.

4 Split third-party-only vs. comprehensive, then apply premiums

Within the insured base, owners either buy the cheapest legally-compliant option (third-party only, price fixed annually by IRDAI) or a full comprehensive/package policy that also covers their own vehicle. That

split should tilt toward comprehensive as vehicle value climbs — nobody buys a bare third-party policy on a ₹3 lakh superbike they financed on EMI.

Segment	TP premium (IRDAI-set)	Comprehensive premium (est.)	Comprehensive share
≤75cc	₹538	~₹1,200	10%
75–150cc	₹714	~₹2,200	35%
150–350cc	₹1,366	~₹5,500	60%
>350cc	₹2,804	~₹14,000	85%

TP rates are the actual IRDAI-notified tariffs for FY26 — worth memorising the rough shape (a few hundred rupees, rising steeply with cc), even if not the exact figures. Comprehensive premiums are reasoned estimates, not published data.

5 Add it up

Segment	GWP (₹ crore)
≤75cc	211
75–150cc	10,781
150–350cc	10,501
>350cc	6,210
Total estimated annual GWP	≈ ₹27,700 crore (~US\$3.3 bn)

6 Sanity check — the step that actually earns the "hire"

A guesstimate nobody checks against reality is just arithmetic performed out loud. Industry research (Mordor Intelligence and comparable trackers) sizes the actual India two-wheeler insurance market at roughly US\$5.66 billion in 2025 — about ₹47,300 crore. Our bottom-up build lands at 59% of that: right order of magnitude, running low by around ₹19,500 crore.

Sanity check: our number vs the real world

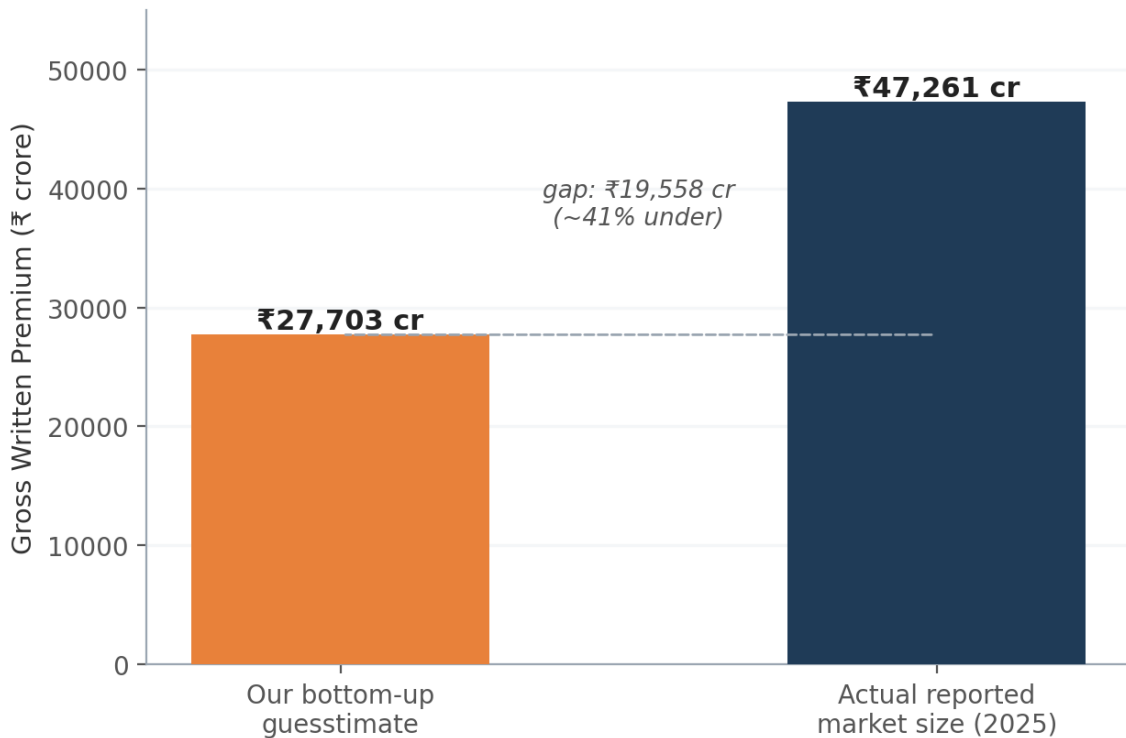


Fig. 2 — Directionally right, materially short. That gap is the interesting part of the answer.

That shortfall isn't a failure — it's the most useful sentence you can say in the room. Three plausible reasons, out loud, in order of likely weight:

1. **Commercial and gig-economy fleets weren't modelled at all.** Food-delivery and last-mile e-commerce two-wheelers are a large, fast-growing, higher-premium segment (insurers price fleet and commercial-use risk higher) that a personal-use-only build simply excludes.
2. **Comprehensive premiums were estimated conservatively.** High-value scooters and electric two-wheelers — with expensive battery packs to insure — carry own-damage premiums well above the round numbers assumed here.
3. **GST and mandatory add-ons inflate written premium** beyond the base policy price: 18% GST plus personal accident cover add-ons sit on top of every number in the table above.

What This Case Is Actually Testing

None of the interviewers asking this question care whether you land on ₹27,700 crore or ₹40,000 crore. They're watching for four things:

- **MECE instinct.** Did you split the market into buckets that don't overlap and cover the whole population, before layering on an assumption — rather than assuming one blended number for 28 crore very different vehicles?
- **Willingness to challenge the "obvious" input.** Most candidates silently assume 100% insurance compliance because the vehicle physically exists. The stronger candidates ask whether that's even true.
- **Comfort narrating uncertainty.** Saying "I don't know the exact penetration rate, so I'll assume X and flag it" out loud is worth more than guessing silently and hoping nobody probes.
- **Actually sanity-checking the output.** Comparing your number to any real-world anchor — even a rough one — and being able to explain a gap instead of being rattled by it.

COMMON MISTAKES TO AVOID

Treating 28 crore two-wheelers as one monolithic block instead of segmenting by a variable that changes behaviour.

Confusing GWP (a yearly flow of premium income) with the total value of vehicles on the road, or with the number of policies sold.

Going silent when unsure of a number — narrate the logic even when the input is a rough guess.

Presenting the final number as precise. "Roughly ₹25,000–30,000 crore, and here's why" reads far better than a suspiciously exact figure.

For whoever presents this to the batch: spend the first five minutes just on Step 3 — cold-calling a few people on "what % of two-wheelers do you think are insured?" before revealing 55–60% uninsured tends to land the whole lesson better than any amount of explanation afterward. Let the surprise do the teaching.

— Consulting Prep Library, Guesstimate Series